

Eco²

WEB / MOBILE · AGENT WORKFLOW · AWS / KUBERNETES

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Backend · Platform · Agent Workflow

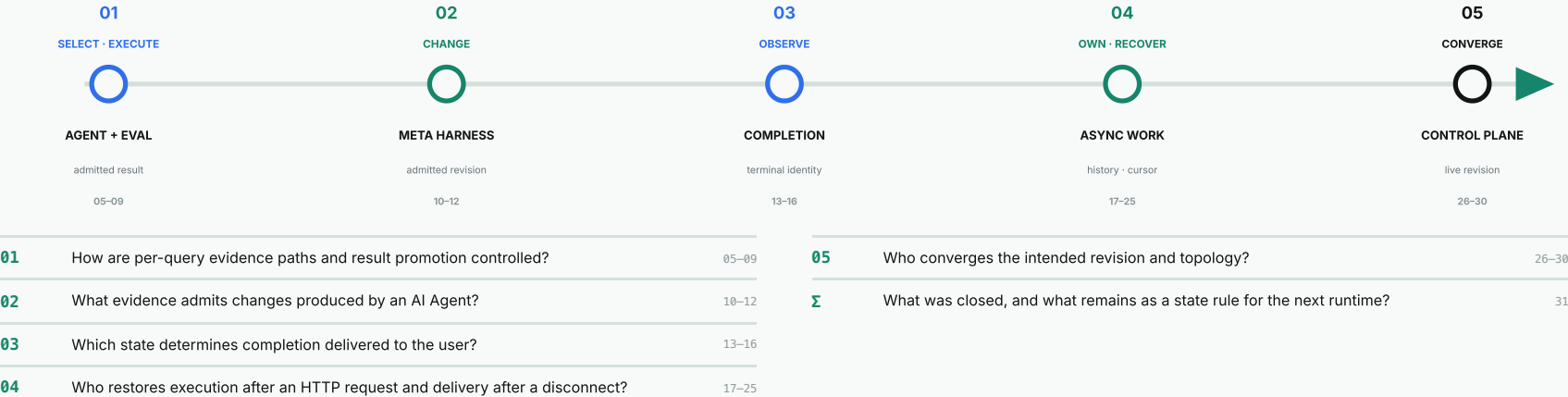
[Portfolio](#) · [Backend](#) · [Frontend](#)

4c3eebb3 / 87c97d61 · 32 PAGES

Contents

The five sections form one design path: execute, change, observe, and recover LLM decisions, then converge platform state.

DECISION SPINE · EACH SECTION HANDS ONE UNRESOLVED OBSERVATION TO THE NEXT OWNER



EVIDENCE CONTRACT

Execution-verification artifact

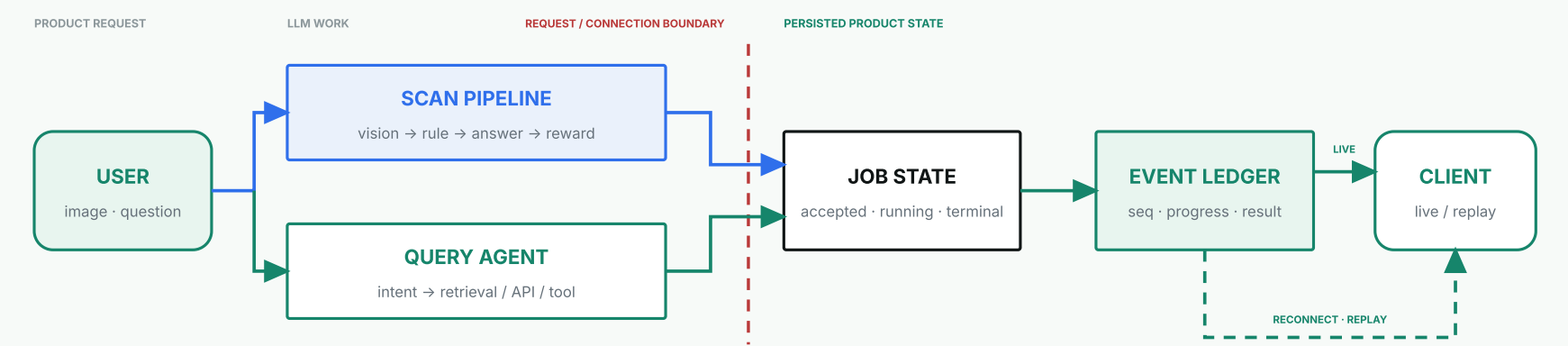
code/test · metric/log/trace · terminal ledger

Promotion authority

human/CI keep-revise-revert · unmeasured remains OPEN

Execution·events·recovery make long-running LLM work a recoverable product state

Scan execution outlived the request, environmental queries branched across retrieval·API·tool paths by intent, and completed results had to be reapplied after SSE disconnects. Acceptance·execution·progress·termination·recovery share one job lifecycle.

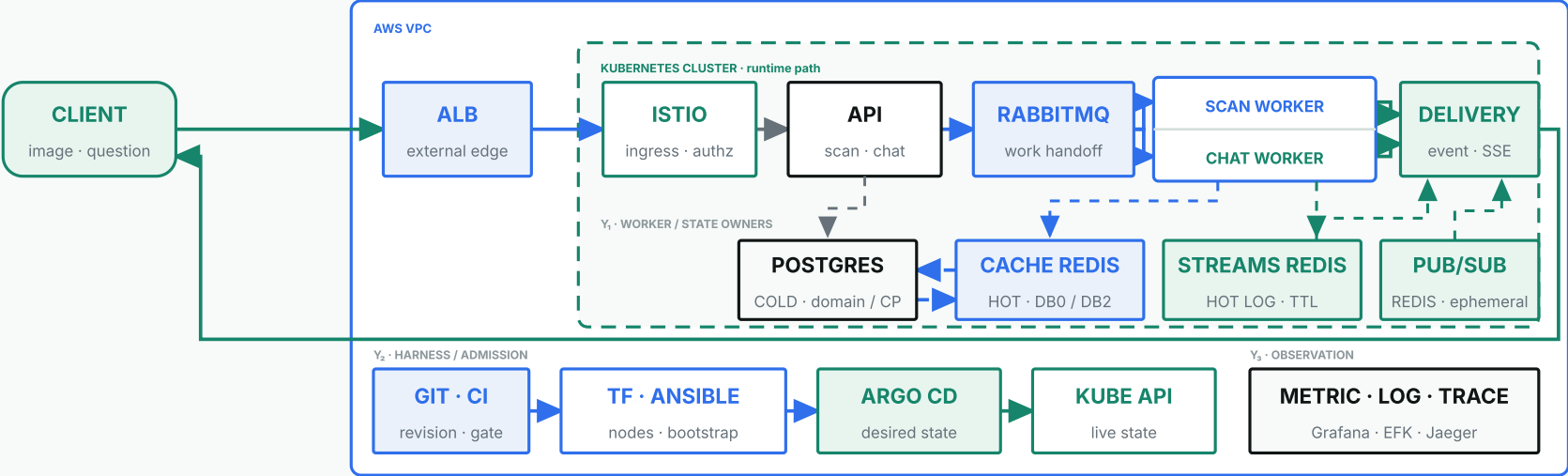


PRODUCT COMPLETION accepted → running → ordered event → client-applied terminal; if the connection closes first, resume from the same event ledger.		
PROBLEM · DECISION QUESTION When a request or SSE connection ends before execution, what state lets the user recover the final result?	EVIDENCE · TEST / RUN ARTIFACT Public product portfolio · Scan application · Chat worker graph · Frontend SSE state	PROMOTION AUTHORITY · OPEN BOUNDARY This design scope is grounded in the current job·event·SSE recovery paths. No per-task impact or recovery success rate is claimed without controlled runs under identical conditions.

Eco² binds LLM decisions to typed execution·state·admission owners

After the API creates an accepted job, RabbitMQ transfers execution ownership to the Scan Celery DAG and Chat LangGraph Agent. The two workers own model/tool execution, checkpoint-retry, and progress-terminal event generation; Event Router→Pub/Sub→SSE delivers them to the user. Cache Redis DB0/DB2 owns hot execution state, PostgreSQL durable domain/cold checkpoints, and Streams Redis TTL event history.

X · REQUEST → WORK HANDOFF → EXECUTION → TERMINAL DELIVERY



PROBLEM · DECISION QUESTION

Columns represent one request lifecycle; rows represent LLM Agent runtime·state·harness/admission·observation owners. Each following section expands one link into its failure and repair contract.

EVIDENCE · TEST / RUN ARTIFACT

[Scan worker queues](#) · [Chat worker task](#) · [Event Router](#) · [Frontend SSE](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

Higher-capability LLMs broaden decisions and tool proposals, but side effect-deployment-completion authority remains with each system owner. Current cluster health-traffic-retention requires separate run evidence.

01

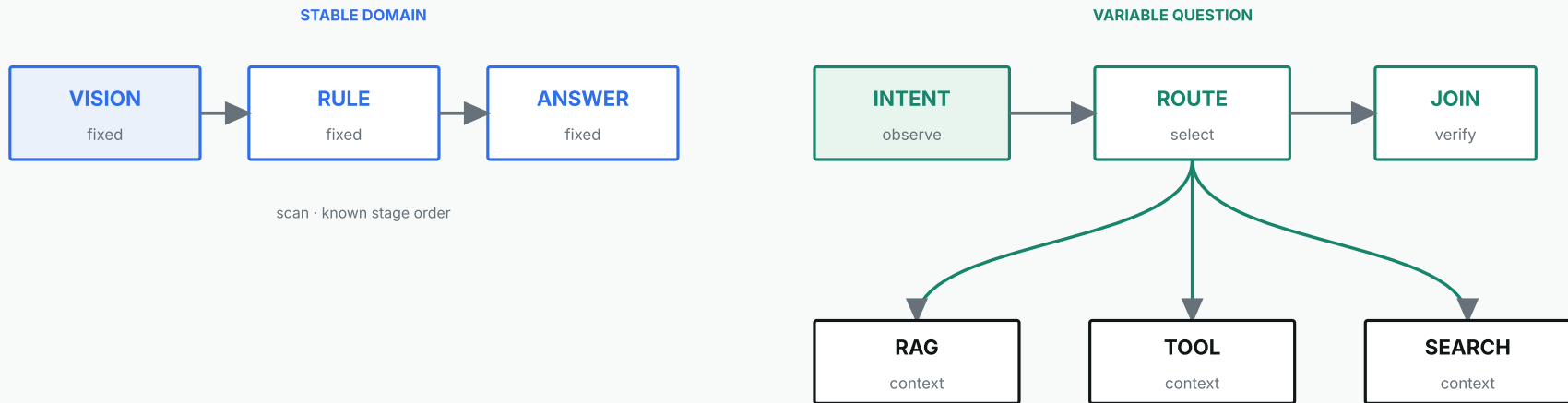
Bind LLM Decisions to Execution Contracts

Separate the fixed Scan DAG from the intent-conditioned query graph, and bind proposal-command-reducer-evaluator authority to execution evidence.

INTENT · TOOL · STATE · REDUCER · EVALUATION

Separate the fixed Scan DAG from the intent-conditioned query graph

Image scan retains a fixed DAG because its input-output schema and stage order remain stable across runs. Environmental queries open conditional routes because intent and tool observations change the retrieval-domain API-search combination. The intent contract fixes required fields with producer-fallback; optional routes preserve the response while recording missing evidence.



INTENT CONTRACT

required · waste / bulk / location / collection point → producer + fallback

OPTIONAL

general / character / weather / web / price / image → fail-open + missing evidence

PROBLEM · DECISION QUESTION

Putting every query in one DAG multiplies unnecessary branches; letting the model own every decision removes the owners of required evidence and fallback. Stable sequences remain fixed stages; paths where new observations change the next operation use an intent-conditioned graph.

EVIDENCE · TEST / RUN ARTIFACT

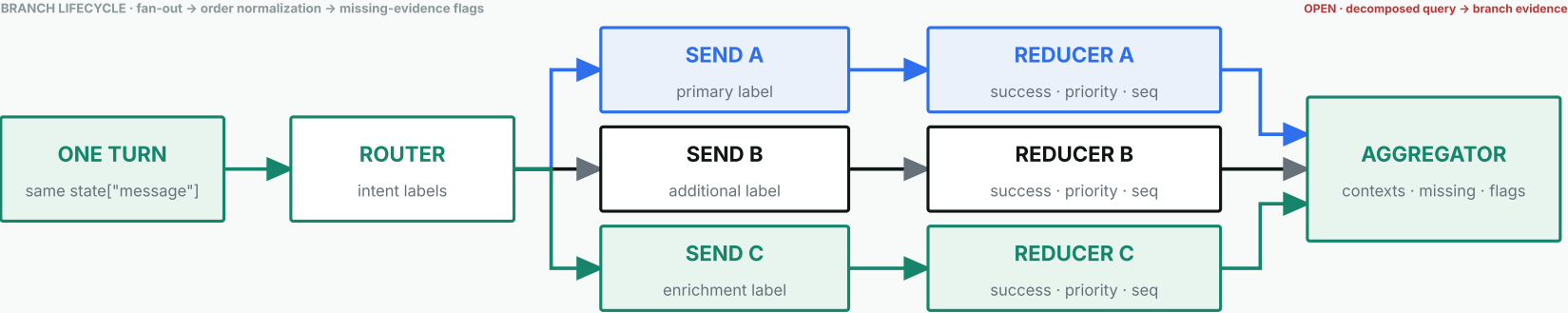
[Scan task chain](#) · [Chat graph factory](#) · [Initial graph decision](#) · [164](#) · [Routing contracts](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

Intent label errors are a new failure surface. Required routes record missing fields as terminal diagnoses; optional routes preserve availability, so missing enrichment must be observed separately.

Reducer order is fixed; query→branch evidence binding remains OPEN

Router fans the same turn state out to the selected intent nodes; `activated_nodes` prevents duplicate execution for the same target. Each channel reducer normalizes nondeterministic arrival order by success-priority-seq, while the aggregator records missing required context and fallback flags. Yet on current main, branch nodes do not read `decomposed_queries/current_query`; they read the same `state[message]`, leaving query-specific evidence binding open.



IMPLEMENTED	<code>activated_nodes</code> admits each target once	OUTPUT	<code>needs_fallback</code> · <code>missing_required</code>
OPEN 1	<code>query_i</code> → <code>branch_i</code> evidence binding	OPEN 2	consumer route for fallback flags
DECISION	arrival timing cannot change reducer order; missing evidence cannot silently become a complete answer		

PROBLEM · DECISION QUESTION

Adding parallel nodes does not determine whether the right evidence is bound to each branch, how late results are merged, or who stops when a required field is missing. Separate admission-reducer-aggregator stages fix each decision point.

EVIDENCE · TEST / RUN ARTIFACT

[Dynamic router](#) · [State reducers](#) · [Aggregator](#) · [Context contracts](#)

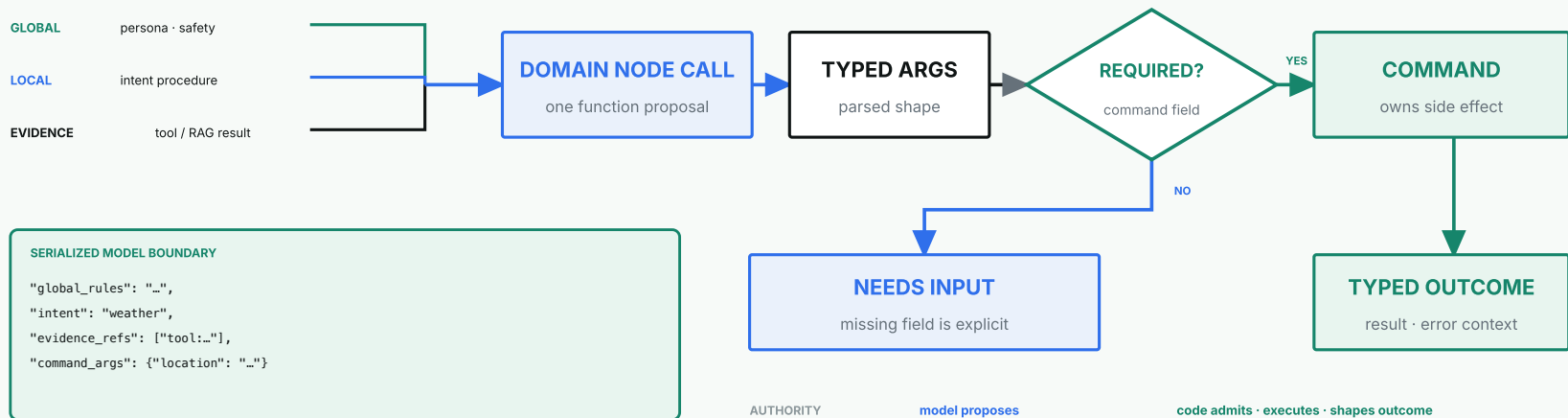
PROMOTION AUTHORITY · OPEN BOUNDARY

Current code proves `Send(node, state)`, node de-duplication, reducer ordering, and output flags. Query-specific branch binding and the fallback consumer route remain OPEN; dynamic replanning is not claimed.

A production domain node executes only a typed command after a one-shot proposal

A route assembles global rules-intent procedure-tool/RAG evidence into one domain-node input. The model proposes only function-call arguments in that node. Missing required fields terminate as an input request; only arguments satisfying the conditions pass through the application command into a typed outcome. This does not mean the entire graph finishes in one model call.

ROUTE-SELECTED CONTEXT → ONE MODEL PROPOSAL → CODE-OWNED SIDE EFFECT



Missing context stops before the side-effect boundary

PROBLEM · DECISION QUESTION

One giant prompt hides route-specific failures; model-owned execution lets plausible arguments obscure execution facts. Separating context selection from side-effect admission recovers which input produced each command.

EVIDENCE · TEST / RUN ARTIFACT

[Prompt loader](#) · [Collection point node](#) · [Weather node](#) · [Answer command](#)

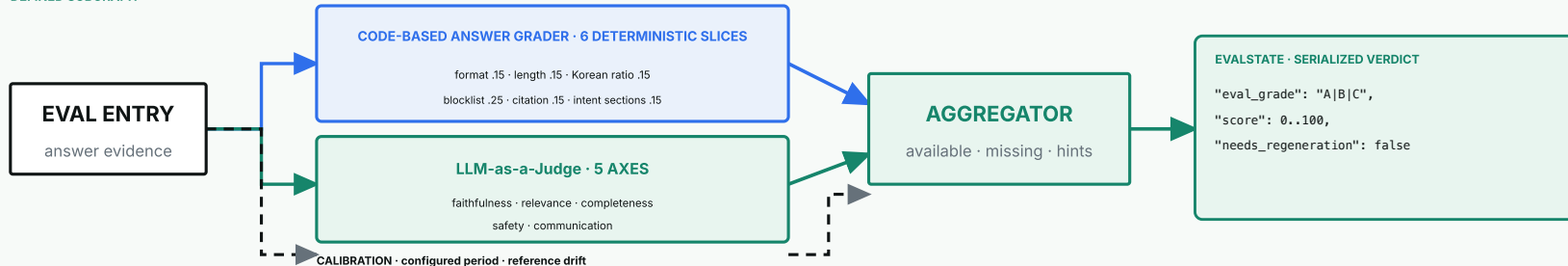
PROMOTION AUTHORITY · OPEN BOUNDARY

This implementation is bounded tool routing. Iterative ReAct loops-dynamic replanning-branch-local query binding are outside the current claim.

Metric schema exists, but missing execution trajectories block promotion

[CodeGraderService](#) serializes format-length-Korean ratio-prohibited expressions-sources-intent sections into 6 fields with explicit weights; unit tests pin each failure. LLM-as-a-Judge defines 5 axes: faithfulness-relevance-completeness-safety-communication. The pinned parent assembly stops at the [eval_counter](#) signature and cannot produce task-level trajectory-metric-review artifacts; this gap blocks promotion.

DEFINED SUBGRAPH



PINNED PARENT ASSEMBLY



NEXT BLOCKERS

missing adapter / asset · parent edge omits counter · intended test cannot collect

Earliest executable failure stops the evidence claim

PROBLEM · DECISION QUESTION

Metrics and reviewer decisions become promotion evidence only with execution trajectories under fixed task-model-workspace-grader conditions. Promotion stops at the first broken executable edge.

EVIDENCE · TEST / RUN ARTIFACT

[Code grader](#) · [Code grader tests](#) · [Eval graph factory](#) · [Dependency assembly](#) · [Parent factory](#) · [ACES](#) · [external eval frame](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

The 6-slice service and unit tests are direct evidence. Production evaluation, paired trials, ATIF normalization, human-grader calibration, regeneration use, and model-quality improvement remain OPEN.

02

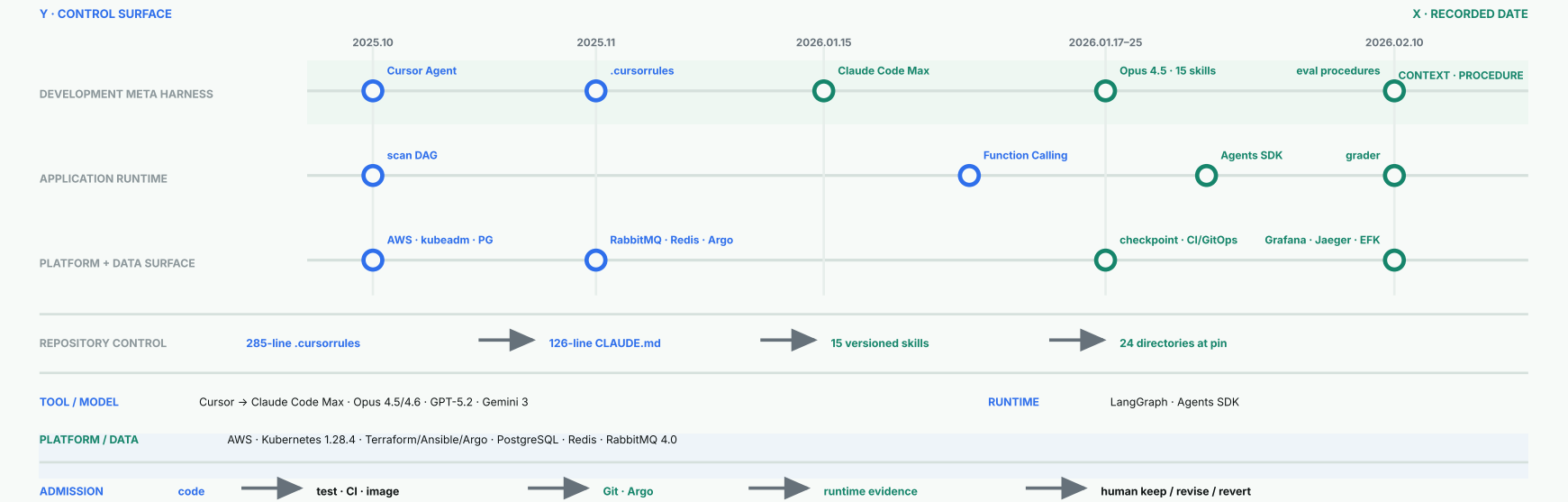
Control Plane for AI Changes

Move conversational instructions into repository procedures; only changes passing code·test·run artifacts advance to the next boundary.

RULE · PROCEDURE · CHANGE · TEST · ADMISSION

Repository procedures admit changes through test·run artifacts·human review

Context once confined to Cursor sessions was decomposed into `.cursorrules`, `CLAUDE.md`, and 15 versioned skills; the current pin contains 24 procedure directories. Procedures provide candidate-change context, path-specific test-image-GitOps-runtime evidence records execution artifacts, and human review decides keep-revise-revert. Reading these three surfaces at the same repository revision constrains the development Agent's change authority.



PROBLEM · DECISION QUESTION

Do procedure descriptions, code/test artifacts, runtime evidence, and reviewer decisions point to the same revision? Only this binding lets the next change reuse the prior decision.

EVIDENCE · TEST / RUN ARTIFACT

[Cursor→Claude Code record](#) · [Opus 4.5 change](#) · [15 versioned skills](#) · [Agents SDK path](#) · [Current runtime deps](#) · [ACES · external eval frame](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

Eco² does not implement ACES paired with/without-skill trials-ATIF-Skill Lift. Skill inventory marginal value, human-grader calibration, defect-rate-token reduction remain unmeasured. Claude skills belong to the development control plane.

Current CI blocks change paths; a dated audit defined the next test scope

Current workflows run Black-Ruff-pytest, Go test, and Terraform-Helm-Kustomize-Kubeconform by changed path, retaining image tags-rendered manifests. A separate 2026-01-14 audit bound unit-test success-error-serialization-connection contracts and pytest-cov-Radon metrics to the same execution record. 115 pass, 49% overall, 88-100% in selected core, and 424 blocks with mean A 2.17 describe that test scope, not current coverage.

X · SELECTOR → VALIDATION → ARTIFACT → ADMISSION

Y · CHANGED SURFACE

CHANGED SURFACE	PATH SELECTOR	STATIC / EXECUTABLE GATE	ARTIFACT	ADMISSION OUTPUT
API SERVICES	apps/{8 APIs} changed-files matrix	black · ruff · pytest format · lint · test	Buildx image revision-bound	tag + manifest next owner
WORKERS	apps/*_worker changed-files matrix	kustomize · black · ruff unit pytest	worker image revision-bound	manifest candidate next owner
SSE / EVENT	gateway · router changed-files matrix	Redis svc · black · ruff pytest	component image revision-bound	realtime candidate next owner
GO AUTHZ	apps/ext_authz changed-files matrix	go test ./... compile · unit	authz image revision-bound	manifest tag next owner
INFRA	tf · charts · overlays changed-files matrix	tf fmt/validate · Helm/ct Kustomize · Kubeconform	rendered manifests revision-bound	GitOps candidate next owner

DATED CHAT_WORKER AUDIT · 2026-01-14

115 / 115
pytest reported pass

49%
total measured coverage

88-100%
selected core paths

A · 2.17
424 blocks · Radon

TEST DESIGN · Port mocks · AAA · fixtures · success/error/serialization/connection

GAP · LLM/LangGraph/setup/presentation → integration/E2E

PROBLEM · DECISION QUESTION The changed-files matrix selects the test environment; executable gates retain artifacts; coverage-complexity metrics and reviewers define the next test scope. Verify that each decision points to the same revision.	EVIDENCE · TEST / RUN ARTIFACT Automatic CI lanes · Dated test-quality audit · Radon / coverage config · Manual Sonar workflow · CodeClimate config	PROMOTION AUTHORITY · OPEN BOUNDARY 115 pass-49%-88-100%-A 2.17 belong to a dated 2026-01-14 artifact. The current pin has no global coverage, paired baseline, live skill lift, or calibrated human score. Radon-CodeClimate-SonarCloud are not automated required gates.
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03

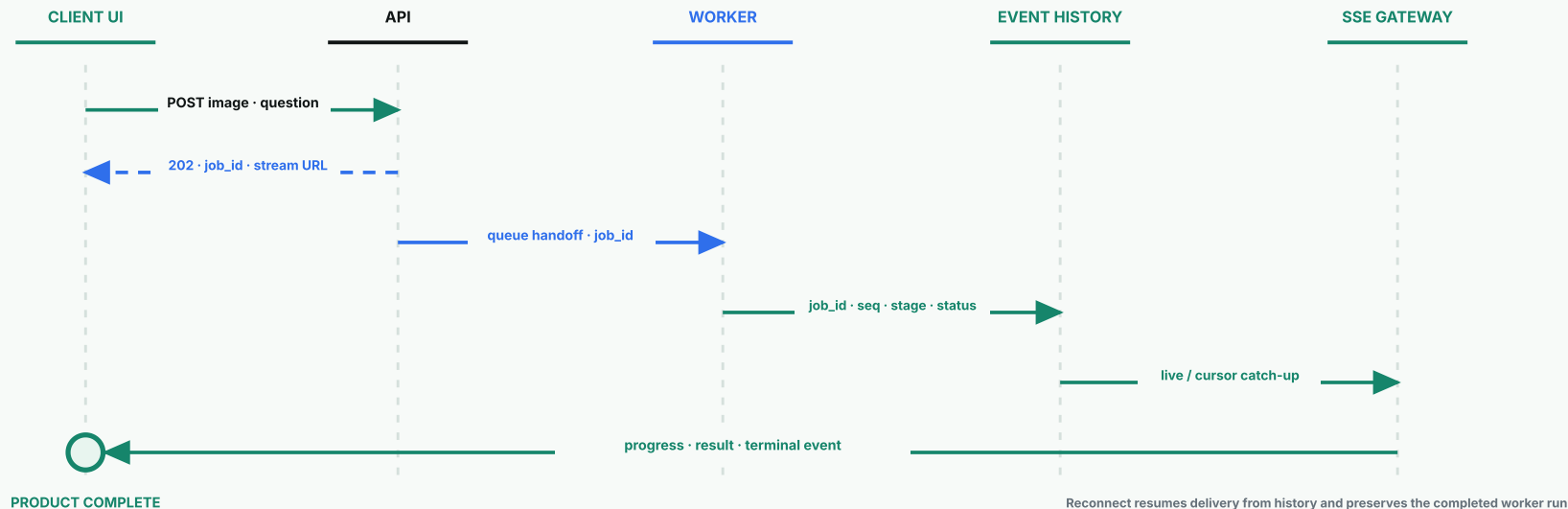
Close Completion at the User Observation

Bind revision-request-terminal state, then determine workflow completion from the state recovered by the client.

REVISION · REQUEST · TERMINAL STATE

Product completion occurs when the client applies the terminal event

API acceptance, worker termination, ordered-event recording, and SSE delivery are distinct observations. A terminal event linked by `job_id · seq` closes this workflow's user result only when live delivery or cursor catch-up applies it to UI state.



PROBLEM · DECISION QUESTION

Separate which API-worker-history-delivery observation can determine user completion. Only client application of the terminal event counts as product completion.

EVIDENCE · TEST / RUN ARTIFACT

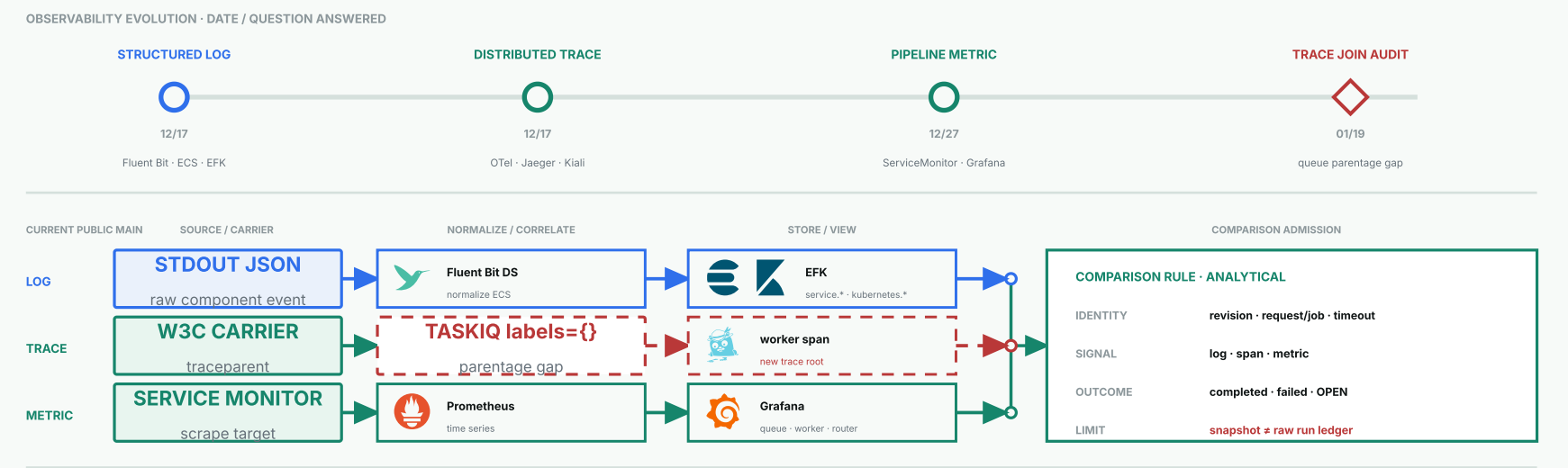
[Scan API contract](#) · [Event Router](#) · [SSE Gateway](#) · [Frontend SSE hook](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

Added event history and catch-up paths enable terminal recovery, while requiring `job_id · seq` retention and reconnect operating cost.

Admit only comparisons binding signal·revision·terminal to the same run

EFK normalizes component logs into ECS fields; Jaeger observes span parentage; Prometheus·Grafana observe queue·worker·router pressure. A dashboard can locate a failure but cannot determine which request under which deployment reached which terminal. Artifact revision, request/job identity, timeout, and terminal denominator therefore define comparison eligibility. This is not a deployed gate; it is the admission criterion for artifact interpretation and human review.



Signal localizes mechanism; only identity + terminal denominator can authorize comparison

PROBLEM · DECISION QUESTION

Component signals alone cannot recover the causal path from deployment change to user outcome. Signals own mechanism localization; the revision·request·job·terminal join owns comparison eligibility.

EVIDENCE · TEST / RUN ARTIFACT

[Fluent Bit · ECS](#) · [Jaeger application](#) · [Event Bus dashboard](#) · [Chat trace handoff](#)

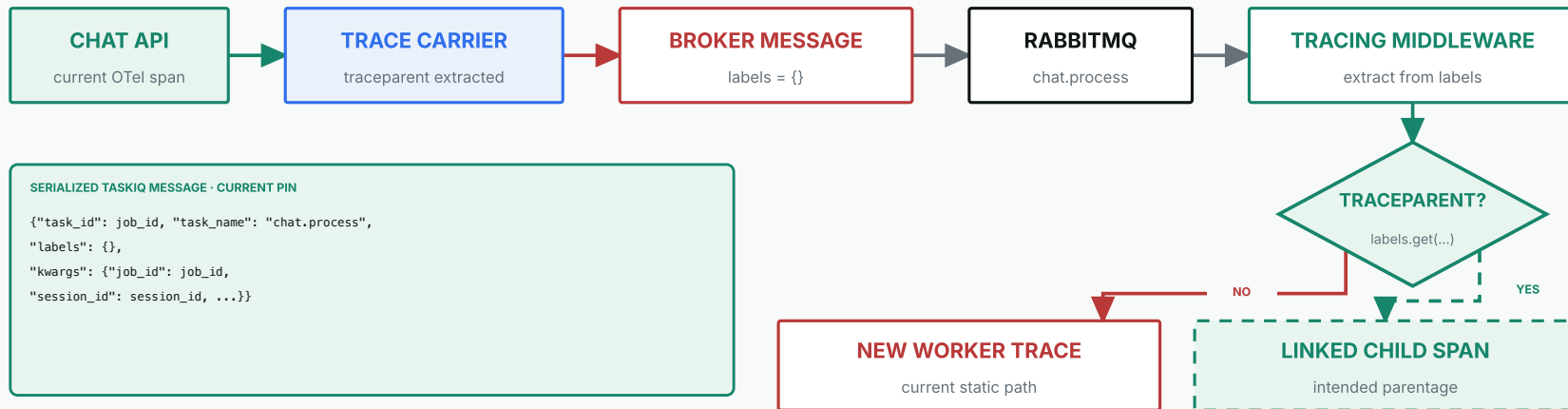
PROMOTION AUTHORITY · OPEN BOUNDARY

Queue labels on pinned main are empty, leaving API→worker parentage open. Jaeger uses memory storage-15,000 trace cap, Elasticsearch is single-node, and Grafana snapshots do not replace the raw k6 ledger.

Empty queue labels break parentage between Chat API and worker traces

Pinned main extracts the W3C carrier but serializes `labels` in both TaskIQ payload and BrokerMessage as `{}`. Without traceparent, worker middleware starts a consumer span without parent context, so current code cannot claim one API→MQ→worker trace chain.

CURRENT PINNED MESSAGE PATH



HISTORICAL LOG AUDIT

125,398 / 1,750,699 = 7.16% with trace.id

window · revision unavailable

PROBLEM · DECISION QUESTION

job_id binds work to Redis events; traceparent binds latency parentage between API and worker spans. Separate identifier authority locates the missing link at the code boundary.

EVIDENCE · TEST / RUN ARTIFACT

[Chat job submitter](#) · [Worker tracing middleware](#) · [Redis event trace fields](#) · [Historical trace audit - 41](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

Current finding is a static code-path audit, not measured runtime incidence. The historical 7.16% ratio lacks an exact window and revision.

04

Recover Execution Beyond the Request Lifetime

Deadline and delivery pressure separate the owners of job-history-client recovery.

JOB IDENTITY · HISTORY · ACK · CURSOR · TERMINAL OUTCOME

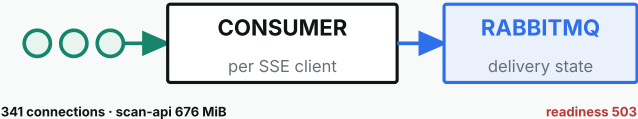
Synchronous timeouts and broker pressure split execution·delivery ownership

Under synchronous Scan, the HTTP request owned execution until Vision·Rule·Answer·Reward completed. When the estimated queue at VU 100 reached about 150 seconds, exceeding the ALB's 60 seconds, the API returned a job reference and RabbitMQ·Celery took over execution. At VU 50, asynchronous v1 exposed one RabbitMQ consumer per client, 341 connections, scan-api at 676 MiB, and readiness 503. The current path assigns task-ordered history·live delivery-recovery to distinct owners.

DATED PRESSURE · TWO OWNERSHIP FAILURES



12/28 · ASYNC V1 · 50 VU



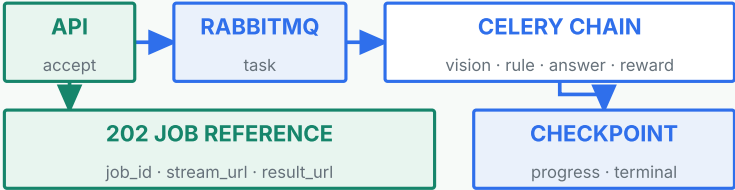
OBSERVATION

execution left HTTP; delivery retained broker state

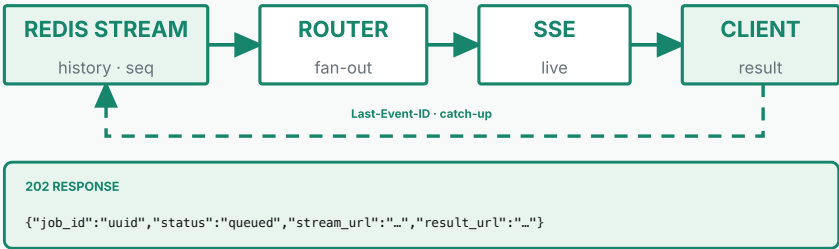
DECISION

split task, history, live delivery and recovery owners

ADOPTED OWNER GRAPH · CURRENT PIN

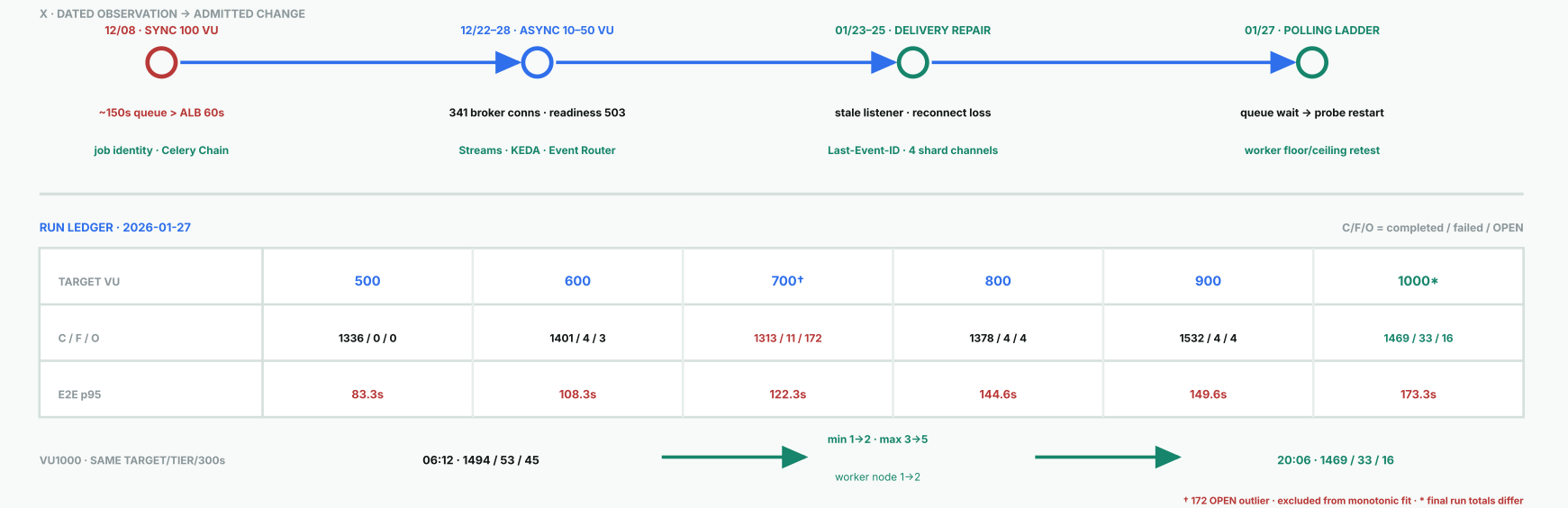


DELIVERY OWNERS · ordered history → fan-out → live → client



Separating failure owners enabled 500→1,000 VU verification

After separating the synchronous request deadline from per-client broker state and adding a reconnect contract, the polling workload expanded across 500-600-700-800-900-1,000 VU. Each run's completed-failed-OPEN and E2E p95 determine the next load investigation. At 1,000 VU, probe restarts remained a bottleneck after worker floor-ceiling changes.



PROBLEM · DECISION QUESTION

Did each workload strengthen the exposed deadline-connection-queue-probe owner before admitting the next verification load?

EVIDENCE · TEST / RUN ARTIFACT

[VU500 report · 252](#) · [VU1000 report · 259](#) · [Current scaling overlay](#) · [Current shard delivery](#)

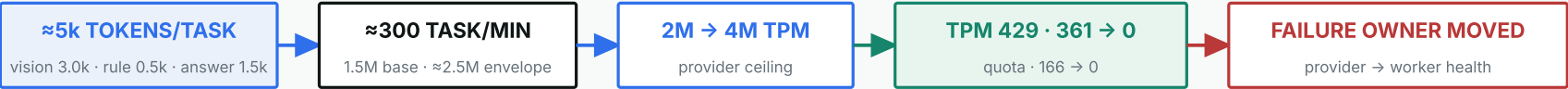
PROMOTION AUTHORITY · OPEN BOUNDARY

VU is k6 verification load, not actual user growth. Runs in month 12 and month 1, including some month 1 runs, differ in config-submitted total; the raw ledger is chronological and does not form one performance curve.

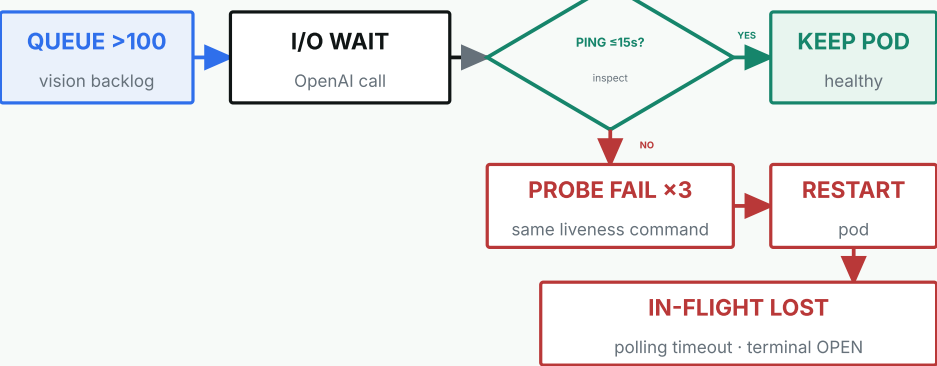
After Tier 4 closed 429s, the retained report traced probe restarts

Demand near 5k tokens/task had exceeded the 2M TPM ceiling; at 4M-auto-recharge, 429 closed at 0 events. In a multi-change VU1000 run, E2E p95 remained 173.3 seconds after increasing worker floor-ceiling and nodes. The retained report identified the next verification target: queue backlog delays `celery inspect ping`, causing liveness using the same command to fail and potentially lose tasks during restart.

PROVIDER BOUNDARY · REPORT ESTIMATE



RETAINED FAILURE CHAIN · FINAL VU1000 REPORT



SAME VU1000 / TIER4 · MULTI-CHANGE OBSERVATION

	06:12 INITIAL	20:06 FINAL
worker min / max	1 / 3	2 / 5
worker nodes	1	2
failed	53	33
throughput · req/min	378.7	373.4
submit p95 · s	0.863	0.787
poll p95 · s	2.547	2.609
E2E p95 · s	166.7	173.3
E2E p99	OPEN	OPEN

PROBLEM · DECISION QUESTION

Provider ceiling owns 429s, KEDA-node count owns worker supply, and liveness owns restarts. Separating failure owners leaves independent HTTP health and queue backpressure as the next verification targets.

EVIDENCE · TEST / RUN ARTIFACT

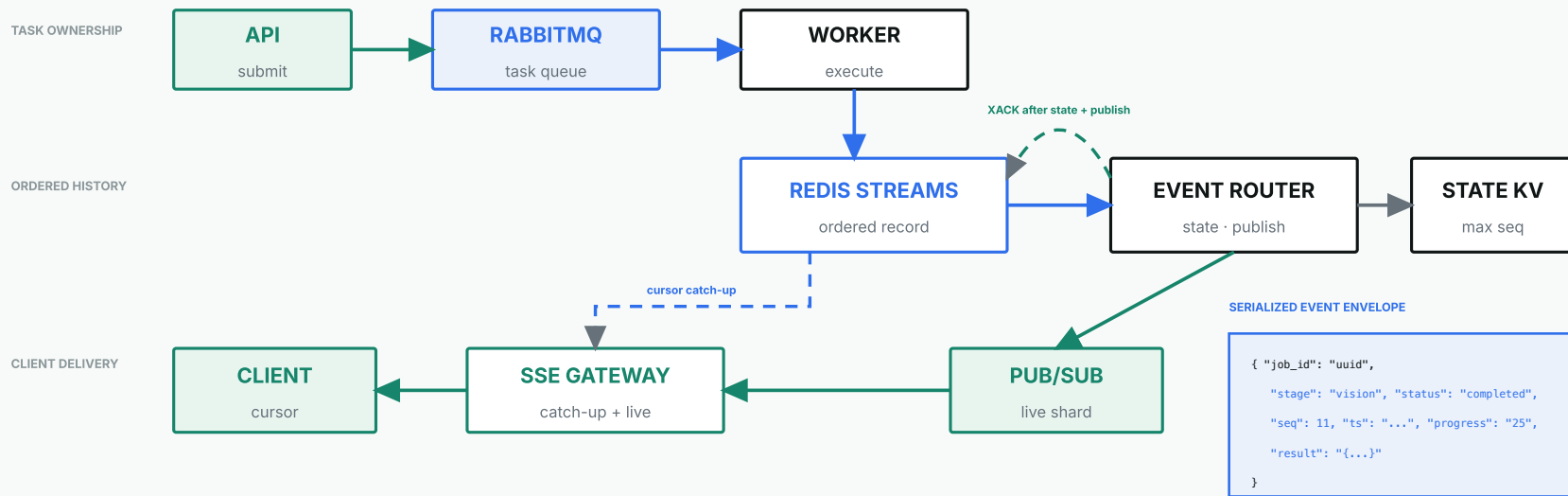
[Tier 3 report](#) · [Tier 4 report](#) · [Final run report](#) · [Current worker probes](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

The two Tier 4 runs differ in submitted total and worker config, so no causal claim is made for the 53→33 decrease. The probe chain and task loss are retained-report diagnoses; independent `/health` remains recommended-unimplemented. E2E p99 is absent from the report and remains OPEN.

Bind task·history·live delivery·recovery with job_id·seq

Responsibility splits from the initial RabbitMQ task handoff into ordered Streams history, Pub/Sub live fan-out, and PEL-Last-Event-ID recovery. Workers serialize stage events as Streams records; the Router XACKs only after state update and Pub/Sub publish succeed. The SSE Gateway finishes cursor catch-up before joining the live shard.



PROBLEM · DECISION QUESTION

Do job_id·seq in each Streams record trace ACK, state update, catch-up, and live resume through one event lifecycle?

EVIDENCE · TEST / RUN ARTIFACT

[Event Router source](#) · [SSE Gateway source](#) · [Stage event serializer](#) · [HA architecture - 100](#)

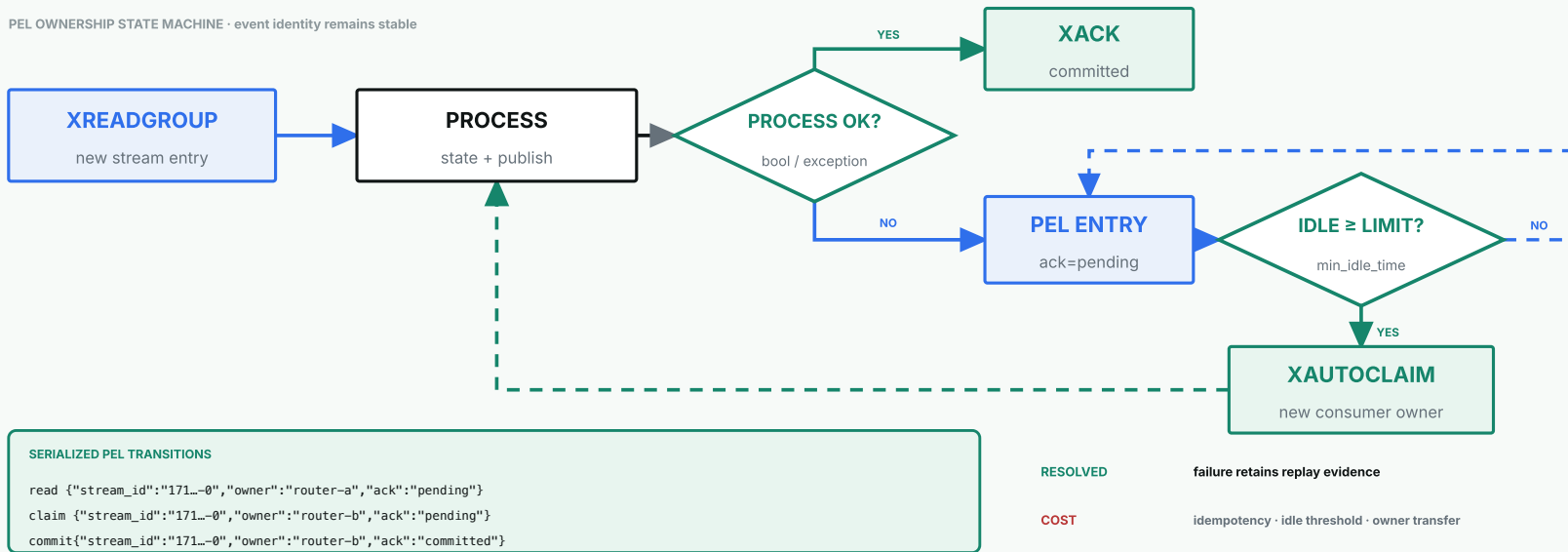
PROMOTION AUTHORITY · OPEN BOUNDARY

Logical responsibility separation is shown; current public Streams storage is not claimed as long-term durable.

Failed entries remain in PEL; XAUTOCLAIM transfers only processing ownership

When `process_event` returns false or raises an exception, XACK is skipped and the same `stream_id` stays in PEL. XAUTOCLAIM transfers entries past the idle threshold to a new consumer; reprocessing closes as committed only after state update and publish succeed. Event identity remains fixed while owner and acknowledgement state transition.

PEL OWNERSHIP STATE MACHINE · event identity remains stable



PROBLEM · DECISION QUESTION

ACKing a processing failure erases replay evidence; copying the original splits attempt identity. Keeping the same stream entry pending and transferring only ownership preserves failure-safe replay and idempotency decisions.

EVIDENCE · TEST / RUN ARTIFACT

[Consumer success-before-XACK](#) · [XAUTOCLAIM reclaimer](#) · [Integrity repair](#) · [237](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

Current source proves pending, reclaim, and success-before-XACK branches. Idle thresholds and consumer identity require operation; no pinned matched replay benchmark exists.

Place independent queue snapshots beside terminal counts, not causal comparisons

Independent Grafana snapshots at 600-800-900-1,000 VU locate pending-active maxima and router p99. A separate dated run report retains 1,518 submissions: 1,469 completed, 33 failed, 16 OPEN. Without shared revision-window identity, queue values are not interpreted as causing outcomes or trending with VU.



PROBLEM · DECISION QUESTION

Grafana owns the location of pending-active-router latency; the k6 report owns the submitted-completed-failed-OPEN denominator. Place them together, but draw no causal arrow without shared identity.

EVIDENCE · TEST / RUN ARTIFACT

[VU600 snapshot](#) · [VU800 snapshot](#) · [VU900 snapshot](#) · [VU1000 snapshot](#) · [VU1000 run report](#) · 259

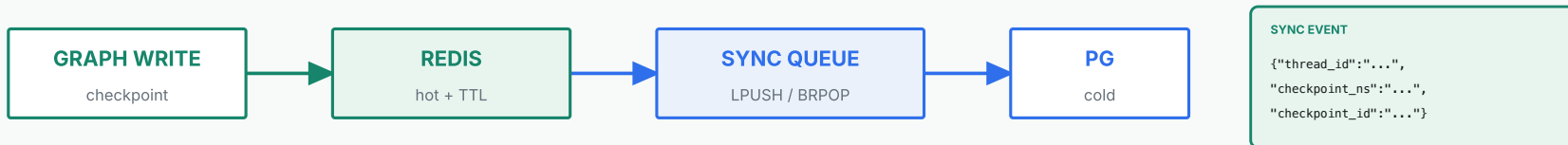
PROMOTION AUTHORITY · OPEN BOUNDARY

Queue bars are independent per-snapshot maxima on a shared count axis; router p99 is a millisecond annotation. 96.8% uses all submissions; 97.8% uses 1,502 resolved cases. Matched time-series and causal before/after remain unmeasured.

Synchronize Redis checkpoints to PostgreSQL for cold recovery

Writes leave Redis state and a sync event; reads load a PostgreSQL checkpoint and promote it to Redis only on a hot miss. Three identities in the sync event bind the cold copy to the same graph state.

WRITE PATH



READ PATH



checkpoint recovery and realtime event recovery retain separate cursors

PROBLEM · DECISION QUESTION

Redis owns hot writes, the sync queue owns async persistence, and PostgreSQL owns cold recovery.

EVIDENCE · TEST / RUN ARTIFACT

[SyncableRedisSaver](#) · [Checkpoint sync service](#) · [Read-through checkpointer](#) · [Pool incident · 242](#)

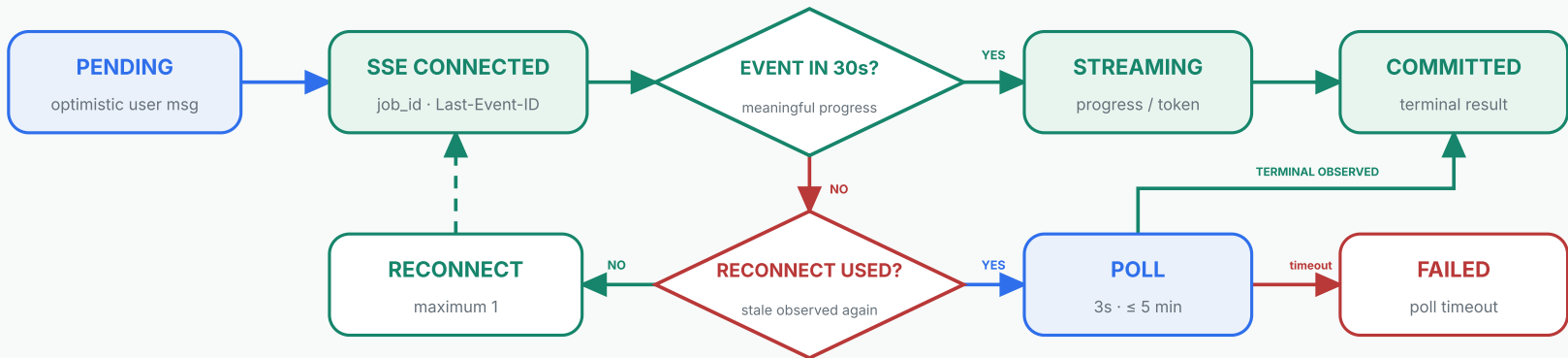
PROMOTION AUTHORITY · OPEN BOUNDARY

Current wiring and the cited artifacts support this decision. Subsequent state-operating cost-outcomes require separate runs.

Reconnect and polling are recovery actions that reobserve terminal state

The client explicitly transitions pending→streaming→committed→failed. Last-Event-ID catch-up retrieves unseen Redis Stream entries. With no meaningful event in 30 seconds, it reconnects once; if still stale, it polls every 3 seconds for up to 5 minutes. Only a terminal observed by polling transitions to committed; timeout closes as failed.

CLIENT MESSAGE STATE MACHINE · predicate labels own every branch



INDEXEDDB · persisted reconciliation state

```
{ "client_id": "...", "session_id": "...", "last_event_id": "171-0", "status": "committed", "synced": 1 }
```

COST

SSE + polling transport · local/server identity reconciliation

PROBLEM · DECISION QUESTION

Track whether reconnect and polling obtain new observations before client state closes as committed or failed.

EVIDENCE · TEST / RUN ARTIFACT

[SSE request parsing](#) · [Catch-up implementation](#) · [IndexedDB session hierarchy](#) · [Agent SSE recovery](#) · [Stale polling fallback](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

Information-gathering actions absorb transient gaps but add latency-API traffic-local/server identity reconciliation costs.

05

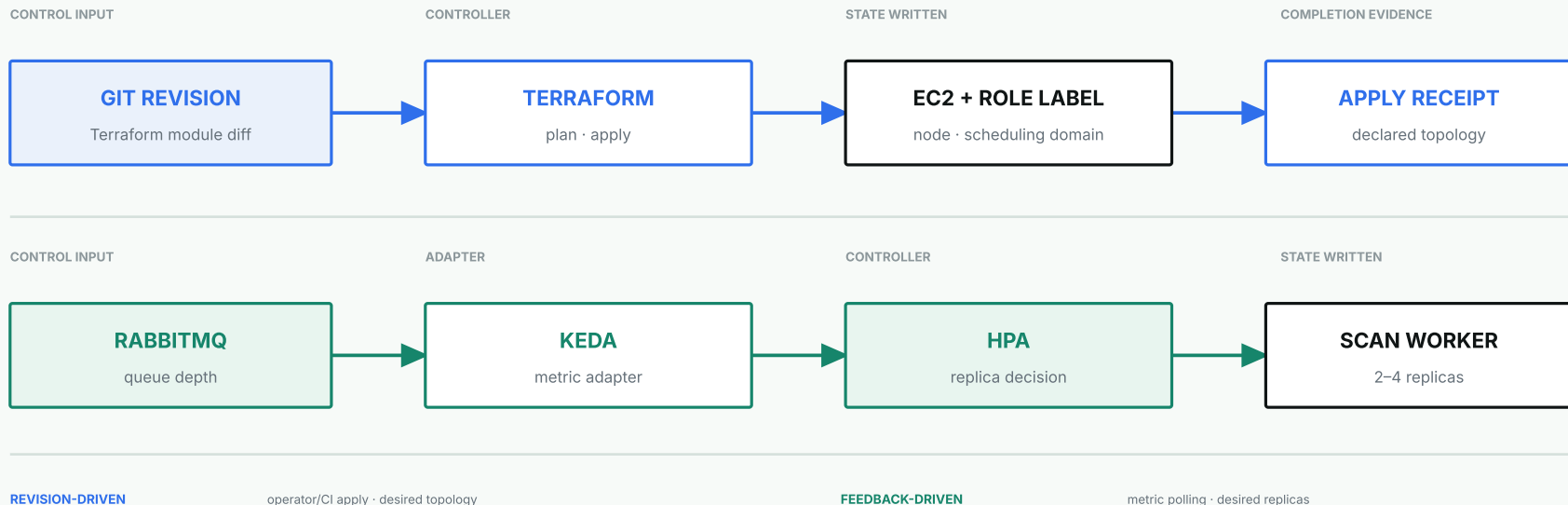
Platform Control Plane

Resource-specific controllers converge revision and topology.

PROVISION · BOOTSTRAP · RECONCILE · SCALE · ROUTE

Git revision changes node existence; queue depth changes worker replicas

Git revision→Terraform declares EC2 and role labels, closing at the apply receipt. RabbitMQ queue depth→KEDA→HPA adjusts desired scan-worker replicas within 2–4. The two flows remain separate axes because inputs-controllers-recorded states differ.



PROBLEM · DECISION QUESTION

Which controller owns each change: physical resource existence or event-driven workload replicas?

EVIDENCE · TEST / RUN ARTIFACT

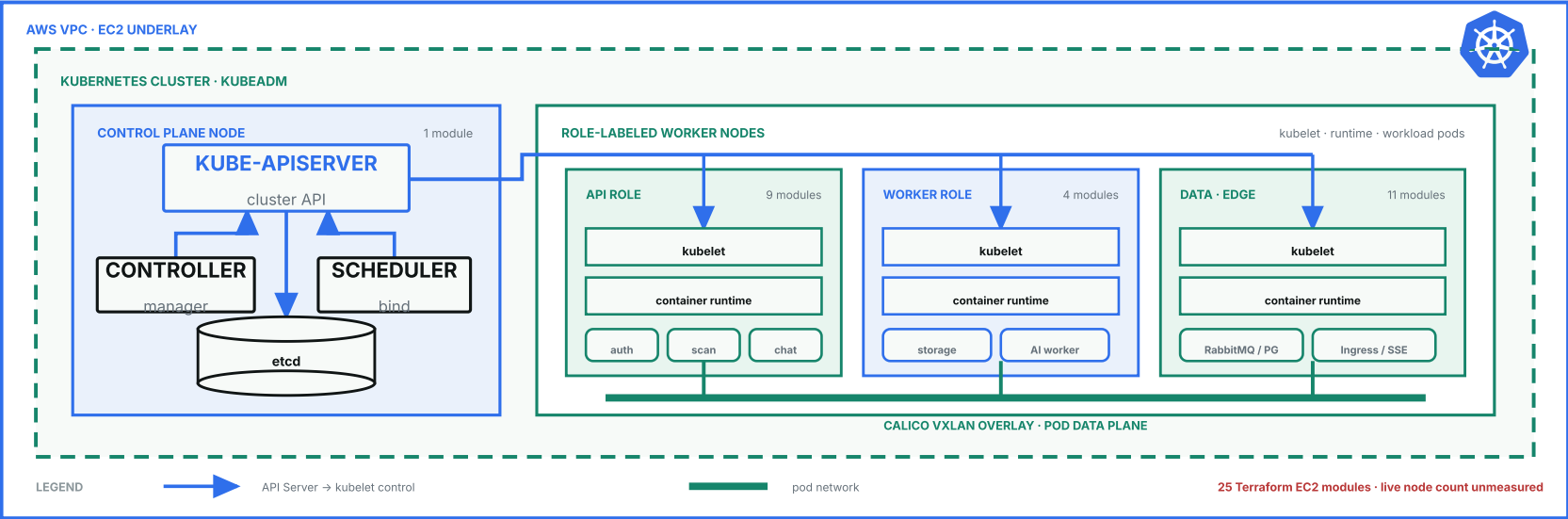
[Terraform current](#) · [Scan worker ScaledObject](#) · [KEDA RabbitMQ scaler](#) · [KEDA scaling model](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

Public desired-state wiring only; live node inventory and realized autoscaling outcomes are not measured here.

Terraform role labels separate Kubernetes control paths and failure domains

An AWS VPC contains the kubeadm cluster; within it, control-plane nodes and three worker role groups are separated. Kube-apiserver coordinates cluster state with controller-scheduler-etcd and sends control requests to each node's kubelet. Pod traffic uses the Calico VXLAN data plane.



PROBLEM · DECISION QUESTION

Bind control-plane state, kubelet control traffic, and pod data traffic to distinct boundaries, fixing the first investigation point for scheduling-node-network failures.

EVIDENCE · TEST / RUN ARTIFACT

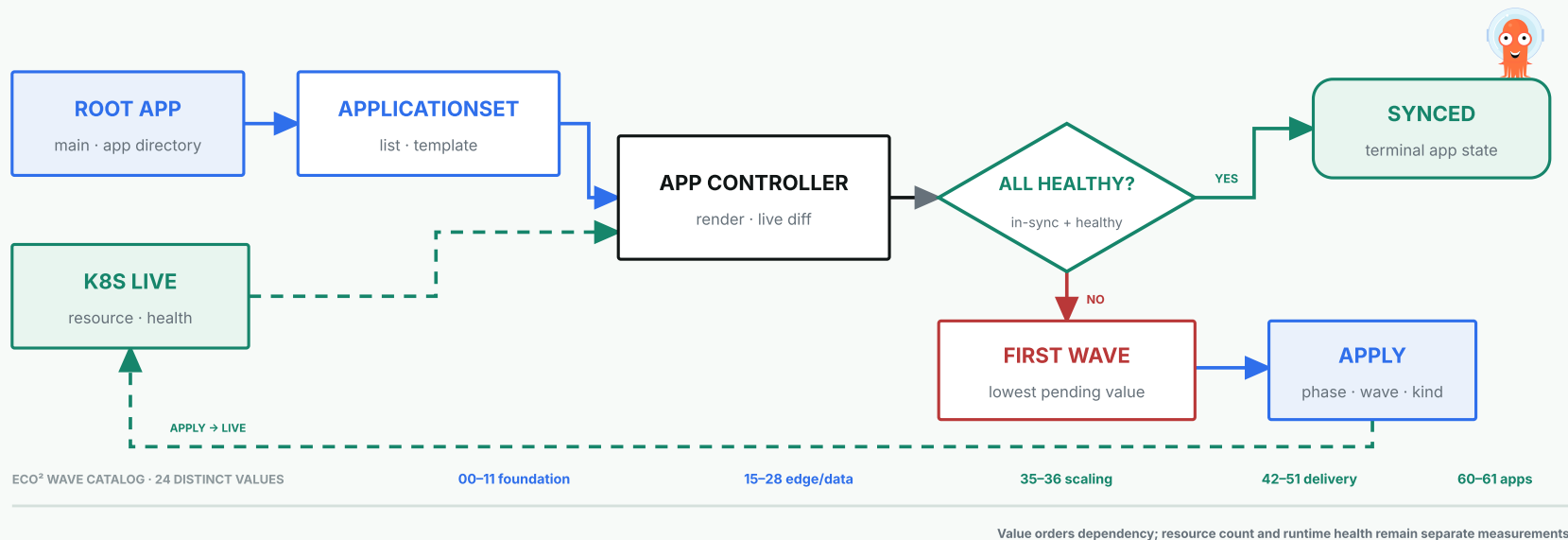
[Terraform main](#) · [Kubelet role labels](#) · [Kubernetes components](#) · [Public portfolio](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

Role labels provide a code-declared scheduling boundary. Operating three worker groups and 25 Terraform modules costs more; live node inventory and realized isolation outcomes are not measured here.

Root App converges expanded Applications through per-wave health loops

`prod-root` reads main's app directory, and ApplicationSet creates child Application specs. The App controller compares rendered desired state with Kubernetes live resources, applies the lowest pending wave, then rereads health. The 24 values encode dependency order, not resource count or performance.



PROBLEM · DECISION QUESTION

Separate Git path discovery·Application creation·live reconciliation·health decisions into distinct control phases, then repeat at the first wave with an unmet prerequisite.

EVIDENCE · TEST / RUN ARTIFACT

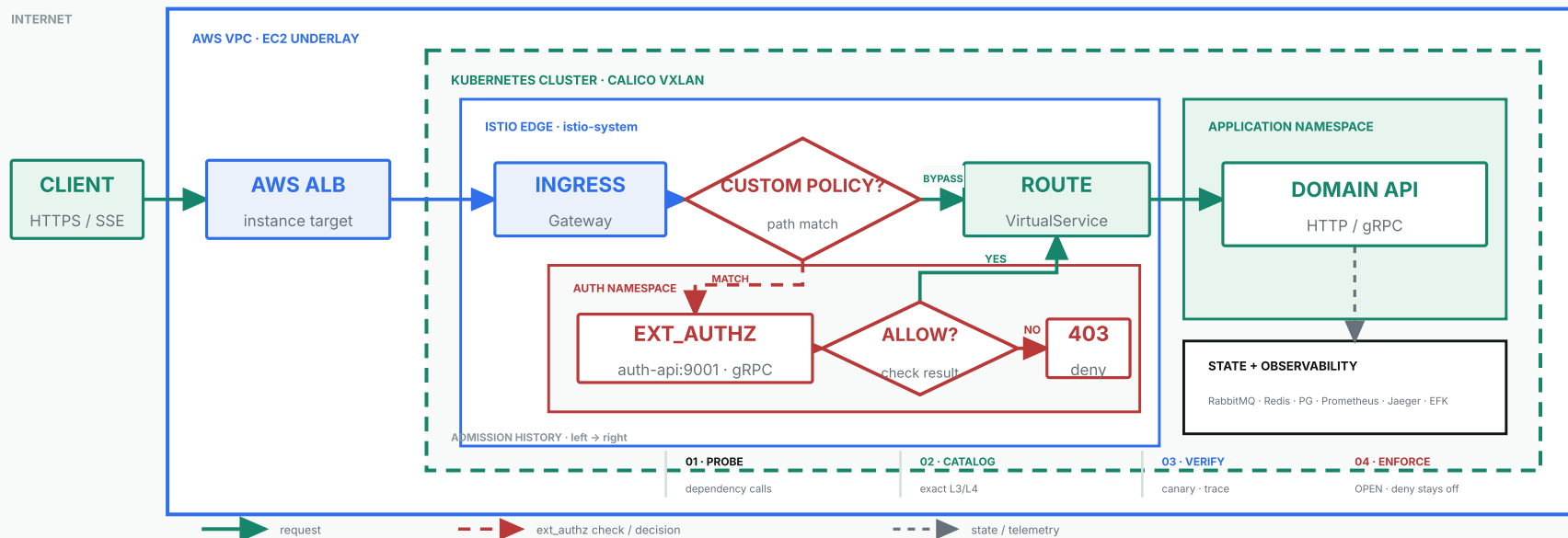
[Production Argo CD apps](#) · [Infrastructure validation](#) · [Argo sync waves](#) · [CNCF · configuration drift](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

Recursive convergence repairs drift but requires operating the child catalog and sync-wave spacing together. Moving a value changes dependency and rollout order, so retain only actual prerequisites.

The allow catalog is refined; default-deny enforcement remains deferred

Client requests pass through ALB-Istio ingress; only CUSTOM policy targets receive ext-authz decisions. A separate default-deny probe exposed internal call paths including DNS-DB-gRPC-event-telemetry, and confirmed dependencies refined the L3/L4 allow catalog. Without closed live canary-packet traces, full default-deny remains disabled.



PROBLEM · DECISION QUESTION

Reachability, external authorization, and internal dependencies have distinct owners and failures. One policy would blur the causes of 403s and broken communication, so admission follows observation→exact rule→canary/trace.

EVIDENCE · TEST / RUN ARTIFACT

[Istio Gateway](#) · [CUSTOM AuthorizationPolicy](#) · [Extension provider](#) · [Istio custom authz](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

The figure shows the current manifest's desired route. Actual packet path-latency, FQDN enforcement, and full default-deny remain OPEN; they become enforcement candidates only when that evidence exists.

Eco² recovered async completion; unresolved observations become state rules

Eco² closes execution-history-client terminal-platform controller authority at implementation boundaries. Trace parentage, branch-local evidence, evaluation assembly, and live enforcement remain OPEN at the current pin. Preserving this distinction, an analysis rule transfers to GEODE trajectory admission: when identical observations require different next actions, use paired history to find the minimal separating field.

01

CLOSED IN ECO²

ASYNC HANDOFF

job identity takes over execution after the API deadline

ORDERED RECOVERY

Streams-PEL-Last-Event-ID preserve missing history

CLIENT TERMINAL

reconnect-polling restore committed / failed

CONTROL OWNERS

distinct controllers converge revision-replica-route

02

OPEN AT PIN

TRACE PARENTAGE

queue labels={} · API→worker join incomplete

BRANCH BINDING

decomposed query→branch evidence unbound

EVAL TRAJECTORY

execution assembly stops at parent signature

LIVE ENFORCEMENT

default-deny-p99-matched causal run unmeasured

03

TRANSFER RULE · PROPOSED

Refine state only when the same projection requires different next actions.

h₁ · terminal applied → STOP

h₂ · receipt missing → REPLAY / POLL

Replay paired history under the same contract; add only the minimal field separating actions to the candidate projection.

CANDIDATE · job_id · seq · stream_id · client status

COST · serialization · migration · storage · fallback traffic

PROBLEM · DECISION QUESTION

Do not conflate implementation-closed boundaries, gaps reproduced at the current pin, and analysis rules for the next system.

EVIDENCE · TEST / RUN ARTIFACT

[Pinned backend](#) · [POMDP](#) · [Kaelbling et al.](#) · [State abstraction](#) · [AAAI](#) · [ReAct](#) · [ICLR 2023](#)

PROMOTION AUTHORITY · OPEN BOUNDARY

POMDP-state abstraction-ReAct provide analytical context. Eco² does not implement a POMDP solver-belief state-RL policy-projection-admission loop. The transfer rule is a proposed analytical framing.

References

Implementation questions start from pinned code, performance questions from run artifacts, choice rationale from dated narratives, and conceptual interpretation from external frames. Each section footer links directly to the selected source.

	SOURCE TYPE	PRIMARY REFERENCES	OWNS	DOES NOT PROVE
01	CODE AUTHORITY	Backend pinned tree · Frontend pinned tree · CHANGELOG	class-function-manifest-test wiring	separate runs own live health-traffic
02	RUN ARTIFACT	VU1000 run · Grafana · pre-gevent · Grafana · VU1000 · Test-quality audit	terminal-latency-queue-coverage-complexity	causality across unmatched runs remains OPEN
03	DATED NARRATIVE	Public portfolio · Engineering chronology · Debugging context · Pinned commit	problem observation-choice rationale-change order	historical records do not override current wiring
04	EXTERNAL FRAME	Trustworthy Agents · OpenAI agent guide · ACES · POMDP · State abstraction · ReAct	agent-evaluation-state-refinement vocabulary	external research is not Eco ² implementation evidence

READING ORDER

Code pin → run artifact → human/CI decision · dated narratives and external frames support rationale and interpretation; they do not replace execution facts.